

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Data Structures

Experience

- **Amazon** May 2025 – Aug 2025
Software Development Engineer Intern Seattle, US
 - Built an end-to-end **AI agent** with **RAG** and an **agentic workflow** that retrieves operational logs, reasons over incident context, and generates remediation guidance for production operators.
 - Implemented a **Model Context Protocol (MCP)** server for tool-augmented **LLM** retrieval, cutting query latency and compute cost; provisioned **AWS CDK** on **ECS Fargate** with **multi-stage CI/CD**.
 - Shipped a **React** and **TypeScript** UI integrated with backend **REST APIs**; iterated on prototypes through user feedback, evaluations, and code reviews with senior engineers.
- **CampusMesh** Jan 2026 – Present
AI Software Engineer New York, US
 - Prototyped and productionized **LLM retrieval pipelines (RAG)** for CampusIQ, an **AI assistant** using **embedding-based semantic search** and structured indexes to improve answer quality by **30%**.
 - Developed **Python/Node.js** backend services on **AWS (Lambda, API Gateway, S3)** across **4 environments**; documented API contracts and tooling to accelerate future agent development.
- **New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
AI Software Engineer New York, US
 - Partnered with research and product stakeholders to build full-stack tools (**React, Python, Node.js, REST APIs**) on an NSF-funded platform; gathered feedback and shipped UI changes for scientific and operations users.
 - Integrated **LLM-powered workflows (GPT-4 + LangChain)** for document analysis and reporting; engineered **Python ETL** into **PostgreSQL** and dashboards for experiment-style telemetry review.
 - Deployed **computer vision** models (**YOLOv8, EfficientNet**) and cloud inference pipelines for structured data capture from mobile inputs used in health research workflows.
- **PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer Bengaluru, India
 - Built an **NLP OCR pipeline (LayoutLM)** on **Azure** to automate structured extraction from scientific documents, reducing processing time by **75%**.
 - Maintained production backend services on **Azure** with **SQL Server**; mentored engineers on testing and defect resolution in a regulated environment.

Projects

- **InviLite (AI Agent & Analytics Platform)** | *Python, AWS Bedrock, MCP, FAISS, Lambda, Docker*
 - Built agent-style tooling for ingestion, classification, and triage; integrated **RAG (FAISS + Bedrock LLM)** into an automated analysis and reporting pipeline, lifting classification accuracy from **56% to 92%**.
- **RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face*
 - Designed a **PEFT/LoRA** fine-tuning pipeline for **RoBERTa-base**; built evaluation workflows and achieved **83.45%** accuracy with under **1M** trainable parameters.

Skills

- **Programming Languages:** Python, TypeScript, JavaScript (ES6), Java, SQL, C++
- **AI/ML & Agents:** LLMs, AI Agents, RAG (LangChain, FAISS), GPT-4, Hugging Face, PyTorch, LLM Fine-tuning, MCP, NLP, Computer Vision
- **Web & Backend:** React, Node.js, Express.js, REST API, Microservices
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS (Bedrock, Lambda, ECS, SageMaker, CDK, S3, API Gateway, IAM), Azure, Docker, Kubernetes, CI/CD
- **Tools & Platforms:** Git, Linux, Jupyter, Cursor, GitHub Copilot, Postman

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals